

(a) textbook hard-threshold ABC-PMC: one update per generation

P_{t+1} : the N kept, weighted;
the rejected are discarded

(b) generation-free (ours): one update per arrival

θ^* to a free worker; ρ^* arrives later:
append $(\theta^*, \rho^*, \epsilon_{t^*}, \tau^*, w^*)$, $\tau^* = n$

the current population only; earlier generations are discarded

1
state

next threshold: a quantile of the population's discrepancies; accept or reject

2
tolerance ϵ

weights carried over from the last generation; covariance from them

3
weights, Σ

one mixture proposal over the whole population

4
proposal

propose, simulate, accept or reject, one candidate at a time, until N are kept

5
draw

the kept are weighted against that one proposal, after acceptance

6
weight

every particle ever evaluated, with its stored bandwidth and weight

the k best form the archive; a smooth kernel of width ϵ_n , which only ever tightens

archive weights: stored weight times kernel weight; covariance from them

one mixture proposal over the archive, rebuilt at every call

one candidate: a parent drawn by weight, then perturbed

weighted against the mixture of all proposals used so far, before it is simulated

barrier: q_{t+1} needs the complete P_{t+1}

$|\mathcal{H}_n| < k$: prior draw, $w^* = 1$,
no ϵ stamp, straight to a worker